

Tables for “*On the Uncertainty Quantification Ability of Tabular  
Foundation Models*”

**Table 1 Summary of results:** RRMSE, NIS, and wall-clock time are reported (medians  $\pm$  std across 20 runs). Best numbers are in bold font and noise refers to either  $\varepsilon \sim \mathcal{N}(0, (0.005 \times c)^2)$  or  $\varepsilon \sim \mathcal{N}(0, (0.05 \times c)^2)$  where  $c$  is a problem-dependent constant that adjusts the noise variance based on the problem. All GPs use default GP+ settings. TabPFN runs on GPU (CPU time in parentheses) and the compute time is almost entirely for inference. GP runs on CPU with the majority of its compute time spent during training.

| Problem     | N                | $D_x$ | Noise | GP+                                 |                                     |                                   | TabPFN v2.5                         |                                     |                                   |
|-------------|------------------|-------|-------|-------------------------------------|-------------------------------------|-----------------------------------|-------------------------------------|-------------------------------------|-----------------------------------|
|             |                  |       |       | RRMSE                               | NIS                                 | CPU Time (s)                      | RRMSE                               | NIS                                 | GPU Time (s)                      |
| Wing Weight | 10D <sub>x</sub> | 10    | 0.005 | <b>0.016 <math>\pm</math> 0.002</b> | <b>0.073 <math>\pm</math> 0.022</b> | <b>1.07 <math>\pm</math> 0.13</b> | 0.033 $\pm$ 0.004                   | 0.167 $\pm$ 0.008                   | 1.37 $\pm$ 0.04                   |
|             | 10D <sub>x</sub> | 10    | 0.05  | <b>0.068 <math>\pm</math> 0.004</b> | 0.331 $\pm$ 0.032                   | <b>1.11 <math>\pm</math> 0.14</b> | 0.070 $\pm$ 0.006                   | 0.327 $\pm$ 0.026                   | 1.38 $\pm$ 0.05                   |
|             | 40D <sub>x</sub> | 10    | 0.005 | <b>0.007 <math>\pm</math> 0.000</b> | <b>0.033 <math>\pm</math> 0.001</b> | 7.44 $\pm$ 0.93                   | 0.013 $\pm$ 0.002                   | 0.082 $\pm$ 0.007                   | <b>1.49 <math>\pm</math> 0.02</b> |
|             | 40D <sub>x</sub> | 10    | 0.05  | <b>0.055 <math>\pm</math> 0.001</b> | <b>0.257 <math>\pm</math> 0.005</b> | 7.90 $\pm$ 0.62                   | 0.057 $\pm$ 0.002                   | 0.269 $\pm$ 0.007                   | <b>1.45 <math>\pm</math> 0.26</b> |
| Buckling    | 10D <sub>x</sub> | 4     | 0.005 | <b>0.174 <math>\pm</math> 0.094</b> | 1.567 $\pm$ 1.348                   | 9.64 $\pm$ 2.28                   | 0.419 $\pm$ 0.539                   | 0.656 $\pm$ 0.377                   | <b>1.10 <math>\pm</math> 0.04</b> |
|             | 10D <sub>x</sub> | 4     | 0.05  | <b>0.304 <math>\pm</math> 0.121</b> | 3.384 $\pm$ 1.378                   | 4.97 $\pm$ 1.96                   | 0.323 $\pm$ 0.441                   | <b>1.035 <math>\pm</math> 0.372</b> | <b>1.09 <math>\pm</math> 0.06</b> |
|             | 40D <sub>x</sub> | 4     | 0.005 | <b>0.016 <math>\pm</math> 0.013</b> | <b>0.056 <math>\pm</math> 0.040</b> | 11.88 $\pm$ 3.16                  | 0.097 $\pm$ 0.126                   | 0.155 $\pm$ 0.046                   | <b>1.06 <math>\pm</math> 0.03</b> |
|             | 40D <sub>x</sub> | 4     | 0.05  | <b>0.078 <math>\pm</math> 0.024</b> | <b>0.360 <math>\pm</math> 0.109</b> | 6.42 $\pm$ 2.82                   | 0.231 $\pm$ 0.193                   | 0.469 $\pm$ 0.065                   | <b>1.07 <math>\pm</math> 0.05</b> |
| Ackley      | 10D <sub>x</sub> | 10    | 0.005 | <b>0.301 <math>\pm</math> 0.023</b> | <b>1.569 <math>\pm</math> 0.173</b> | 1.16 $\pm$ 0.16                   | 0.378 $\pm$ 0.042                   | 1.887 $\pm$ 0.192                   | <b>1.15 <math>\pm</math> 0.03</b> |
|             | 10D <sub>x</sub> | 10    | 0.05  | <b>0.310 <math>\pm</math> 0.022</b> | <b>1.616 <math>\pm</math> 0.172</b> | <b>1.13 <math>\pm</math> 0.12</b> | 0.383 $\pm$ 0.041                   | 1.938 $\pm$ 0.194                   | 1.15 $\pm$ 0.03                   |
|             | 40D <sub>x</sub> | 10    | 0.005 | 0.213 $\pm$ 0.004                   | 1.073 $\pm$ 0.027                   | 7.61 $\pm$ 0.79                   | 0.210 $\pm$ 0.005                   | <b>0.993 <math>\pm</math> 0.018</b> | <b>1.27 <math>\pm</math> 0.03</b> |
|             | 40D <sub>x</sub> | 10    | 0.05  | 0.220 $\pm$ 0.004                   | 1.112 $\pm$ 0.027                   | 7.88 $\pm$ 0.69                   | 0.217 $\pm$ 0.005                   | 1.036 $\pm$ 0.018                   | <b>1.28 <math>\pm</math> 0.03</b> |
|             | 10D <sub>x</sub> | 20    | 0.005 | <b>0.316 <math>\pm</math> 0.013</b> | <b>1.614 <math>\pm</math> 0.079</b> | 3.37 $\pm$ 0.36                   | 0.385 $\pm$ 0.025                   | 1.830 $\pm$ 0.115                   | <b>1.75 <math>\pm</math> 0.01</b> |
|             | 10D <sub>x</sub> | 20    | 0.05  | <b>0.321 <math>\pm</math> 0.014</b> | <b>1.621 <math>\pm</math> 0.091</b> | 3.42 $\pm$ 0.34                   | 0.390 $\pm$ 0.025                   | 1.866 $\pm$ 0.113                   | <b>1.75 <math>\pm</math> 0.26</b> |
|             | 40D <sub>x</sub> | 20    | 0.005 | 0.217 $\pm$ 0.003                   | 1.069 $\pm$ 0.014                   | 59.54 $\pm$ 8.38                  | 0.214 $\pm$ 0.005                   | 0.995 $\pm$ 0.022                   | <b>2.15 <math>\pm</math> 0.01</b> |
|             | 40D <sub>x</sub> | 20    | 0.05  | 0.225 $\pm$ 0.003                   | 1.096 $\pm$ 0.014                   | 60.42 $\pm$ 9.44                  | <b>0.222 <math>\pm</math> 0.005</b> | <b>1.037 <math>\pm</math> 0.022</b> | <b>2.15 <math>\pm</math> 0.02</b> |
|             | 10D <sub>x</sub> | 40    | 0.005 | <b>0.298 <math>\pm</math> 0.011</b> | <b>1.495 <math>\pm</math> 0.047</b> | 18.00 $\pm$ 3.23                  | 0.416 $\pm$ 0.023                   | 1.962 $\pm$ 0.074                   | <b>3.02 <math>\pm</math> 0.01</b> |
|             | 10D <sub>x</sub> | 40    | 0.05  | <b>0.307 <math>\pm</math> 0.017</b> | <b>1.531 <math>\pm</math> 0.080</b> | 17.91 $\pm$ 3.13                  | 0.421 $\pm$ 0.022                   | 1.987 $\pm$ 0.075                   | <b>3.02 <math>\pm</math> 0.01</b> |
|             | 40D <sub>x</sub> | 40    | 0.005 | 0.243 $\pm$ 0.007                   | 1.182 $\pm$ 0.042                   | 467.44 $\pm$ 66.95                | <b>0.230 <math>\pm</math> 0.005</b> | 1.114 $\pm$ 0.016                   | <b>4.65 <math>\pm</math> 0.26</b> |
|             | 40D <sub>x</sub> | 40    | 0.05  | 0.251 $\pm$ 0.006                   | 1.202 $\pm$ 0.035                   | 479.80 $\pm$ 89.74                | <b>0.237 <math>\pm</math> 0.005</b> | 1.138 $\pm$ 0.018                   | <b>4.65 <math>\pm</math> 0.01</b> |
| Dixon Price | 10D <sub>x</sub> | 10    | 0.005 | 0.442 $\pm$ 0.028                   | 2.037 $\pm$ 0.263                   | 1.24 $\pm$ 0.18                   | <b>0.427 <math>\pm</math> 0.072</b> | <b>2.010 <math>\pm</math> 0.318</b> | <b>1.19 <math>\pm</math> 0.03</b> |
|             | 10D <sub>x</sub> | 10    | 0.05  | 0.453 $\pm$ 0.051                   | 2.085 $\pm$ 0.497                   | 1.25 $\pm$ 0.14                   | <b>0.431 <math>\pm</math> 0.069</b> | 2.057 $\pm$ 0.312                   | <b>1.20 <math>\pm</math> 0.03</b> |
|             | 40D <sub>x</sub> | 10    | 0.005 | 0.279 $\pm$ 0.019                   | 1.287 $\pm$ 0.099                   | 11.46 $\pm$ 1.04                  | <b>0.162 <math>\pm</math> 0.018</b> | 0.840 $\pm$ 0.077                   | <b>1.29 <math>\pm</math> 0.03</b> |
|             | 40D <sub>x</sub> | 10    | 0.05  | 0.288 $\pm$ 0.018                   | 1.325 $\pm$ 0.102                   | 11.40 $\pm$ 1.05                  | <b>0.177 <math>\pm</math> 0.017</b> | 0.873 $\pm$ 0.076                   | <b>1.29 <math>\pm</math> 0.02</b> |
|             | 10D <sub>x</sub> | 20    | 0.005 | 0.474 $\pm$ 0.027                   | 2.286 $\pm$ 0.274                   | 4.78 $\pm$ 0.67                   | <b>0.469 <math>\pm</math> 0.028</b> | <b>2.159 <math>\pm</math> 0.128</b> | <b>1.75 <math>\pm</math> 0.25</b> |
|             | 10D <sub>x</sub> | 20    | 0.05  | <b>0.466 <math>\pm</math> 0.025</b> | 2.227 $\pm$ 0.272                   | 4.45 $\pm$ 0.52                   | 0.471 $\pm$ 0.029                   | <b>2.172 <math>\pm</math> 0.131</b> | <b>1.75 <math>\pm</math> 0.01</b> |
|             | 40D <sub>x</sub> | 20    | 0.005 | 0.334 $\pm$ 0.006                   | 1.558 $\pm$ 0.031                   | 99.93 $\pm$ 9.24                  | <b>0.195 <math>\pm</math> 0.009</b> | 0.977 $\pm$ 0.049                   | <b>2.16 <math>\pm</math> 0.02</b> |
|             | 40D <sub>x</sub> | 20    | 0.05  | 0.339 $\pm$ 0.005                   | 1.561 $\pm$ 0.031                   | 99.77 $\pm$ 8.98                  | <b>0.203 <math>\pm</math> 0.009</b> | 1.006 $\pm$ 0.047                   | <b>2.15 <math>\pm</math> 0.01</b> |
|             | 10D <sub>x</sub> | 40    | 0.005 | <b>0.455 <math>\pm</math> 0.025</b> | <b>2.183 <math>\pm</math> 0.153</b> | 31.05 $\pm$ 5.22                  | 0.480 $\pm$ 0.033                   | 2.327 $\pm$ 0.134                   | <b>3.02 <math>\pm</math> 0.01</b> |
|             | 10D <sub>x</sub> | 40    | 0.05  | <b>0.470 <math>\pm</math> 0.022</b> | <b>2.240 <math>\pm</math> 0.135</b> | 32.35 $\pm$ 5.38                  | 0.482 $\pm$ 0.032                   | 2.330 $\pm$ 0.134                   | <b>3.02 <math>\pm</math> 0.01</b> |
|             | 40D <sub>x</sub> | 40    | 0.005 | 0.431 $\pm$ 0.047                   | 2.086 $\pm$ 0.336                   | 867.45 $\pm$ 132.11               | 0.231 $\pm$ 0.009                   | 1.141 $\pm$ 0.034                   | <b>4.65 <math>\pm</math> 0.27</b> |
|             | 40D <sub>x</sub> | 40    | 0.05  | 0.433 $\pm$ 0.044                   | 2.040 $\pm$ 0.300                   | 874.93 $\pm$ 107.68               | 0.237 $\pm$ 0.010                   | 1.163 $\pm$ 0.036                   | <b>4.65 <math>\pm</math> 0.27</b> |
| Griewank    | 10D <sub>x</sub> | 10    | 0.005 | <b>0.011 <math>\pm</math> 0.051</b> | <b>0.111 <math>\pm</math> 0.388</b> | 1.22 $\pm$ 0.12                   | 0.512 $\pm$ 0.071                   | 2.460 $\pm$ 0.313                   | <b>1.15 <math>\pm</math> 0.01</b> |
|             | 10D <sub>x</sub> | 10    | 0.05  | <b>0.109 <math>\pm</math> 0.097</b> | <b>0.529 <math>\pm</math> 0.590</b> | <b>1.13 <math>\pm</math> 0.14</b> | 0.524 $\pm$ 0.069                   | 2.490 $\pm$ 0.317                   | 1.15 $\pm$ 0.01                   |
|             | 40D <sub>x</sub> | 10    | 0.005 | <b>0.007 <math>\pm</math> 0.000</b> | <b>0.033 <math>\pm</math> 0.001</b> | 12.01 $\pm$ 1.05                  | 0.100 $\pm$ 0.006                   | 0.469 $\pm$ 0.028                   | <b>1.27 <math>\pm</math> 0.01</b> |
|             | 40D <sub>x</sub> | 10    | 0.05  | <b>0.056 <math>\pm</math> 0.001</b> | <b>0.262 <math>\pm</math> 0.003</b> | 11.68 $\pm$ 0.89                  | 0.115 $\pm$ 0.006                   | 0.535 $\pm$ 0.028                   | <b>1.27 <math>\pm</math> 0.01</b> |
|             | 10D <sub>x</sub> | 20    | 0.005 | <b>0.139 <math>\pm</math> 0.020</b> | <b>1.009 <math>\pm</math> 0.008</b> | 3.77 $\pm$ 0.29                   | 0.515 $\pm$ 0.073                   | 2.426 $\pm$ 0.329                   | <b>1.76 <math>\pm</math> 0.14</b> |
|             | 10D <sub>x</sub> | 20    | 0.05  | <b>0.174 <math>\pm</math> 0.020</b> | <b>1.065 <math>\pm</math> 0.020</b> | 3.80 $\pm$ 0.39                   | 0.516 $\pm$ 0.072                   | 2.436 $\pm$ 0.327                   | <b>1.76 <math>\pm</math> 0.01</b> |
|             | 40D <sub>x</sub> | 20    | 0.005 | <b>0.008 <math>\pm</math> 0.000</b> | <b>0.040 <math>\pm</math> 0.001</b> | 88.19 $\pm$ 8.96                  | 0.140 $\pm$ 0.006                   | 0.644 $\pm$ 0.022                   | <b>2.16 <math>\pm</math> 0.01</b> |
|             | 40D <sub>x</sub> | 20    | 0.05  | <b>0.061 <math>\pm</math> 0.001</b> | <b>0.282 <math>\pm</math> 0.006</b> | 91.18 $\pm$ 10.18                 | 0.149 $\pm$ 0.006                   | 0.691 $\pm$ 0.024                   | <b>2.16 <math>\pm</math> 0.01</b> |
|             | 10D <sub>x</sub> | 40    | 0.005 | <b>0.255 <math>\pm</math> 0.045</b> | <b>1.552 <math>\pm</math> 0.068</b> | 23.33 $\pm$ 3.28                  | 0.548 $\pm$ 0.043                   | 2.549 $\pm$ 0.172                   | <b>3.02 <math>\pm</math> 0.01</b> |
|             | 10D <sub>x</sub> | 40    | 0.05  | <b>0.257 <math>\pm</math> 0.038</b> | <b>1.562 <math>\pm</math> 0.064</b> | 23.07 $\pm$ 3.50                  | 0.550 $\pm$ 0.045                   | 2.585 $\pm$ 0.177                   | <b>3.03 <math>\pm</math> 0.01</b> |
|             | 40D <sub>x</sub> | 40    | 0.005 | <b>0.008 <math>\pm</math> 0.000</b> | <b>0.047 <math>\pm</math> 0.001</b> | 824.65 $\pm$ 190.64               | 0.173 $\pm$ 0.005                   | 0.970 $\pm$ 0.028                   | <b>4.65 <math>\pm</math> 0.10</b> |
|             | 40D <sub>x</sub> | 40    | 0.05  | <b>0.074 <math>\pm</math> 0.002</b> | <b>0.347 <math>\pm</math> 0.011</b> | 837.12 $\pm$ 194.54               | 0.181 $\pm$ 0.005                   | 1.007 $\pm$ 0.029                   | <b>4.65 <math>\pm</math> 0.27</b> |
| Rosenbrock  | 10D <sub>x</sub> | 10    | 0.005 | 0.459 $\pm$ 0.035                   | 2.206 $\pm$ 0.317                   | <b>1.11 <math>\pm</math> 0.18</b> | 0.379 $\pm$ 0.073                   | <b>2.084 <math>\pm</math> 0.242</b> | 1.18 $\pm$ 0.03                   |
|             | 10D <sub>x</sub> | 10    | 0.05  | 0.462 $\pm$ 0.041                   | 2.285 $\pm$ 0.368                   | <b>1.17 <math>\pm</math> 0.17</b> | 0.386 $\pm$ 0.073                   | 2.079 $\pm$ 0.243                   | 1.17 $\pm$ 0.03                   |
|             | 40D <sub>x</sub> | 10    | 0.005 | 0.337 $\pm$ 0.016                   | 1.588 $\pm$ 0.076                   | 8.70 $\pm$ 0.95                   | <b>0.105 <math>\pm</math> 0.007</b> | 0.500 $\pm$ 0.016                   | <b>1.27 <math>\pm</math> 0.03</b> |
|             | 40D <sub>x</sub> | 10    | 0.05  | 0.340 $\pm$ 0.012                   | 1.617 $\pm$ 0.054                   | 8.74 $\pm$ 1.22                   | 0.122 $\pm$ 0.006                   | 0.567 $\pm$ 0.020                   | <b>1.28 <math>\pm</math> 0.03</b> |
|             | 10D <sub>x</sub> | 20    | 0.005 | 0.490 $\pm$ 0.018                   | 2.435 $\pm$ 0.212                   | 3.99 $\pm$ 0.59                   | <b>0.468 <math>\pm</math> 0.033</b> | <b>2.289 <math>\pm</math> 0.112</b> | <b>1.75 <math>\pm</math> 0.02</b> |
|             | 10D <sub>x</sub> | 20    | 0.05  | 0.492 $\pm$ 0.014                   | 2.395 $\pm$ 0.172                   | 3.93 $\pm$ 0.63                   | <b>0.475 <math>\pm</math> 0.033</b> | 2.302 $\pm$ 0.106                   | <b>1.96 <math>\pm</math> 0.06</b> |
|             | 40D <sub>x</sub> | 20    | 0.005 | 0.373 $\pm$ 0.005                   | 1.752 $\pm$ 0.027                   | 70.29 $\pm$ 7.56                  | <b>0.157 <math>\pm</math> 0.007</b> | <b>0.764 <math>\pm</math> 0.024</b> | <b>2.38 <math>\pm</math> 0.07</b> |
|             | 40D <sub>x</sub> | 20    | 0.05  | 0.377 $\pm$ 0.005                   | 1.774 $\pm$ 0.029                   | 66.62 $\pm$ 10.14                 | <b>0.165 <math>\pm</math> 0.006</b> | <b>0.803 <math>\pm</math> 0.026</b> | <b>2.33 <math>\pm</math> 0.04</b> |
|             | 10D <sub>x</sub> | 40    | 0.005 | <b>0.481 <math>\pm</math> 0.015</b> | <b>2.281 <math>\pm</math> 0.128</b> | 21.93 $\pm$ 4.80                  | 0.501 $\pm$ 0.025                   | 2.371 $\pm$ 0.086                   | <b>3.02 <math>\pm</math> 0.01</b> |
|             | 10D <sub>x</sub> | 40    | 0.05  | <b>0.484 <math>\pm</math> 0.015</b> | <b>2.297 <math>\pm</math> 0.127</b> | 21.65 $\pm$ 3.94                  | 0.503 $\pm$ 0.025                   | 2.406 $\pm$ 0.088                   | <b>3.03 <math>\pm</math> 0.15</b> |
|             | 40D <sub>x</sub> | 40    | 0.005 | 0.422 $\pm$ 0.017                   | 1.997 $\pm$ 0.134                   | 592.45 $\pm$ 104.12               | <b>0.215 <math>\pm</math> 0.006</b> | 1.071 $\pm$ 0.028                   | <b>4.65 <math>\pm</math> 0.01</b> |
|             | 40D <sub>x</sub> | 40    | 0.05  | 0.427 $\pm$ 0.014                   | 2.028 $\pm$ 0.114                   | 578.19 $\pm$ 76.50                | <b>0.222 <math>\pm</math> 0.007</b> | <b>1.099 <math>\pm</math> 0.029</b> | <b>4.65 <math>\pm</math> 0.01</b> |

**Table 2** PE Kernel Dixon-Price and Rosenbrock Summary Statistics: RRMSE, NIS, and wall-clock time (medians  $\pm$  std across 20 runs) are reported for GP (on CPU) and TabPFN (on GPU) across the variations of each problem. All GPs are trained with default settings of GP+ package. Best models for a given problem (lowest median per metric) are in bold.

| Problem     | N                                     | $D_{2\epsilon}$ | Noise | GP+               |                   |                     | TabPFN v2.5       |                                     |                                   | GP+ (PE)                            |                                     |                       |
|-------------|---------------------------------------|-----------------|-------|-------------------|-------------------|---------------------|-------------------|-------------------------------------|-----------------------------------|-------------------------------------|-------------------------------------|-----------------------|
|             |                                       |                 |       | RRMSE             | NIS               | CPU Time (s)        | RRMSE             | NIS                                 | GPU Time (s)                      | RRMSE                               | NIS                                 | GPU Time (s)          |
| Dixon Price | 10D <sub>2<math>\epsilon</math></sub> | 20              | 0.005 | 0.474 $\pm$ 0.027 | 2.286 $\pm$ 0.274 | 4.78 $\pm$ 0.67     | 0.469 $\pm$ 0.028 | 2.159 $\pm$ 0.128                   | <b>1.75 <math>\pm</math> 0.25</b> | <b>0.171 <math>\pm</math> 0.010</b> | <b>0.812 <math>\pm</math> 0.027</b> | 40.24 $\pm$ 5.28      |
|             | 10D <sub>2<math>\epsilon</math></sub> | 20              | 0.05  | 0.466 $\pm$ 0.025 | 2.227 $\pm$ 0.272 | 4.45 $\pm$ 0.52     | 0.471 $\pm$ 0.029 | 2.172 $\pm$ 0.131                   | <b>1.75 <math>\pm</math> 0.01</b> | <b>0.192 <math>\pm</math> 0.015</b> | <b>0.901 <math>\pm</math> 0.084</b> | 36.49 $\pm$ 3.98      |
|             | 40D <sub>2<math>\epsilon</math></sub> | 20              | 0.005 | 0.334 $\pm$ 0.006 | 1.558 $\pm$ 0.031 | 99.93 $\pm$ 9.24    | 0.195 $\pm$ 0.009 | 0.977 $\pm$ 0.049                   | <b>2.16 <math>\pm</math> 0.02</b> | <b>0.068 <math>\pm</math> 0.002</b> | <b>0.313 <math>\pm</math> 0.011</b> | 669.90 $\pm$ 86.70    |
|             | 40D <sub>2<math>\epsilon</math></sub> | 20              | 0.05  | 0.339 $\pm$ 0.005 | 1.561 $\pm$ 0.031 | 99.77 $\pm$ 8.98    | 0.203 $\pm$ 0.009 | 1.006 $\pm$ 0.047                   | <b>2.15 <math>\pm</math> 0.01</b> | <b>0.103 <math>\pm</math> 0.002</b> | <b>0.477 <math>\pm</math> 0.011</b> | 680.67 $\pm$ 160.86   |
| Rosenbrock  | 10D <sub>2<math>\epsilon</math></sub> | 40              | 0.005 | 0.481 $\pm$ 0.015 | 2.281 $\pm$ 0.128 | 21.93 $\pm$ 4.80    | 0.501 $\pm$ 0.025 | 2.371 $\pm$ 0.086                   | <b>3.02 <math>\pm</math> 0.01</b> | <b>0.198 <math>\pm</math> 0.011</b> | <b>0.933 <math>\pm</math> 0.051</b> | 94.39 $\pm$ 32.14     |
|             | 10D <sub>2<math>\epsilon</math></sub> | 40              | 0.05  | 0.484 $\pm$ 0.015 | 2.297 $\pm$ 0.127 | 21.65 $\pm$ 3.94    | 0.503 $\pm$ 0.025 | 2.406 $\pm$ 0.088                   | <b>3.03 <math>\pm</math> 0.15</b> | <b>0.216 <math>\pm</math> 0.012</b> | <b>1.017 <math>\pm</math> 0.057</b> | 98.32 $\pm$ 37.55     |
|             | 40D <sub>2<math>\epsilon</math></sub> | 40              | 0.005 | 0.422 $\pm$ 0.017 | 1.997 $\pm$ 0.134 | 592.45 $\pm$ 104.12 | 0.215 $\pm$ 0.006 | <b>1.071 <math>\pm</math> 0.028</b> | <b>4.65 <math>\pm</math> 0.01</b> | <b>0.109 <math>\pm</math> 0.002</b> | 1.075 $\pm$ 0.126                   | 2288.76 $\pm$ 1005.82 |
|             | 40D <sub>2<math>\epsilon</math></sub> | 40              | 0.05  | 0.427 $\pm$ 0.014 | 2.028 $\pm$ 0.114 | 578.19 $\pm$ 76.50  | 0.222 $\pm$ 0.007 | <b>1.099 <math>\pm</math> 0.029</b> | <b>4.65 <math>\pm</math> 0.01</b> | <b>0.126 <math>\pm</math> 0.002</b> | 1.386 $\pm$ 0.247                   | 2520.30 $\pm$ 1030.66 |

**Table 3** Summary of results for real-world datasets: RRMSE, NIS, and wall-clock time (medians  $\pm$  std across 20 runs). TabPFN reports GPU time with CPU time in parentheses. GP runs on CPU; TabPFN on GPU. All GPs use default GP+ settings.

| Problem   | N                                     | $D_{2\epsilon}$ | GP+                                 |                                                                             |                    | TabPFN v2.5                         |                                                                             |                                   |
|-----------|---------------------------------------|-----------------|-------------------------------------|-----------------------------------------------------------------------------|--------------------|-------------------------------------|-----------------------------------------------------------------------------|-----------------------------------|
|           |                                       |                 | RRMSE                               | NIS ( $C = A + B$ )                                                         | CPU Time (s)       | RRMSE                               | NIS ( $C = A + B$ )                                                         | GPU Time (s)                      |
| Elevators | 10D <sub>2<math>\epsilon</math></sub> | 18              | 0.381 $\pm$ 0.038                   | 2.060 $\pm$ 0.187 = 1.292 $\pm$ 0.124 + 0.768 $\pm$ 0.248                   | 2.40 $\pm$ 0.25    | <b>0.336 <math>\pm</math> 0.145</b> | <b>1.591 <math>\pm</math> 0.229</b> = 1.274 $\pm$ 0.217 + 0.317 $\pm$ 0.103 | <b>0.97 <math>\pm</math> 0.02</b> |
|           | 20D <sub>2<math>\epsilon</math></sub> | 18              | 0.347 $\pm$ 0.016                   | 1.827 $\pm$ 0.089 = 1.275 $\pm$ 0.082 + 0.552 $\pm$ 0.132                   | 9.43 $\pm$ 1.34    | <b>0.302 <math>\pm</math> 0.041</b> | <b>1.413 <math>\pm</math> 0.061</b> = 1.107 $\pm$ 0.083 + 0.306 $\pm$ 0.078 | <b>1.06 <math>\pm</math> 0.04</b> |
|           | 40D <sub>2<math>\epsilon</math></sub> | 18              | 0.324 $\pm$ 0.006                   | 1.659 $\pm$ 0.024 = 1.245 $\pm$ 0.048 + 0.413 $\pm$ 0.057                   | 43.62 $\pm$ 5.26   | <b>0.288 <math>\pm</math> 0.005</b> | <b>1.327 <math>\pm</math> 0.041</b> = 1.029 $\pm$ 0.040 + 0.297 $\pm$ 0.074 | <b>1.23 <math>\pm</math> 0.05</b> |
| Pumadyn32 | 10D <sub>2<math>\epsilon</math></sub> | 32              | <b>0.206 <math>\pm</math> 0.017</b> | <b>0.971 <math>\pm</math> 0.094</b> = 0.762 $\pm$ 0.060 + 0.209 $\pm$ 0.083 | 13.69 $\pm$ 2.20   | 0.257 $\pm$ 0.003                   | 1.202 $\pm$ 0.024 = 0.952 $\pm$ 0.040 + 0.250 $\pm$ 0.052                   | <b>1.60 <math>\pm</math> 0.07</b> |
|           | 20D <sub>2<math>\epsilon</math></sub> | 32              | <b>0.194 <math>\pm</math> 0.002</b> | <b>0.900 <math>\pm</math> 0.016</b> = 0.736 $\pm$ 0.022 + 0.164 $\pm$ 0.028 | 71.44 $\pm$ 11.55  | 0.252 $\pm$ 0.002                   | 1.162 $\pm$ 0.015 = 0.919 $\pm$ 0.023 + 0.243 $\pm$ 0.032                   | <b>1.89 <math>\pm</math> 0.27</b> |
|           | 40D <sub>2<math>\epsilon</math></sub> | 32              | <b>0.186 <math>\pm</math> 0.003</b> | <b>0.950 <math>\pm</math> 0.547</b> = 0.865 $\pm$ 0.028 + 0.085 $\pm$ 0.570 | 423.25 $\pm$ 76.71 | 0.250 $\pm$ 0.005                   | 1.137 $\pm$ 0.030 = 0.907 $\pm$ 0.016 + 0.231 $\pm$ 0.037                   | <b>2.47 <math>\pm</math> 0.01</b> |

**Table 4** Student’s t-Distribution Noise Summary Statistics: RRMSE, NIS, and wall-clock time (medians  $\pm$  std across 20 runs) are reported for GP (on CPU) and TabPFN (on GPU) across the variations of each problem. All GPs are trained with default settings of GP+ package. Best models for a given problem (lowest median per metric) are in bold.

| Problem     | N                | $D_x$ | Noise | GP+                                 |                                                                             |                                   | TabPFN v2.5                         |                                                                             |                                   |
|-------------|------------------|-------|-------|-------------------------------------|-----------------------------------------------------------------------------|-----------------------------------|-------------------------------------|-----------------------------------------------------------------------------|-----------------------------------|
|             |                  |       |       | RRMSE                               | NIS ( $C = A + B$ )                                                         | CPU Time (s)                      | RRMSE                               | NIS ( $C = A + B$ )                                                         | GPU Time (s)                      |
| Wing Weight | 10D <sub>x</sub> | 10    | 0.005 | <b>0.016 <math>\pm</math> 0.002</b> | <b>0.073 <math>\pm</math> 0.022</b> = 0.050 $\pm$ 0.008 + 0.023 $\pm$ 0.027 | <b>1.10 <math>\pm</math> 0.14</b> | 0.036 $\pm$ 0.005                   | 0.173 $\pm$ 0.011 = 0.168 $\pm$ 0.011 + 0.005 $\pm$ 0.004                   | 1.36 $\pm$ 0.04                   |
|             | 10D <sub>x</sub> | 10    | 0.05  | <b>0.068 <math>\pm</math> 0.004</b> | <b>0.331 <math>\pm</math> 0.032</b> = 0.252 $\pm$ 0.018 + 0.079 $\pm$ 0.044 | <b>1.14 <math>\pm</math> 0.14</b> | 0.070 $\pm$ 0.004                   | 0.345 $\pm$ 0.014 = 0.284 $\pm$ 0.022 + 0.061 $\pm$ 0.021                   | 1.37 $\pm$ 0.05                   |
|             | 40D <sub>x</sub> | 10    | 0.005 | <b>0.007 <math>\pm</math> 0.000</b> | <b>0.033 <math>\pm</math> 0.001</b> = 0.026 $\pm$ 0.001 + 0.007 $\pm$ 0.002 | 7.51 $\pm$ 0.96                   | 0.014 $\pm$ 0.003                   | 0.086 $\pm$ 0.008 = 0.086 $\pm$ 0.008 + 0.001 $\pm$ 0.001                   | <b>1.51 <math>\pm</math> 0.06</b> |
|             | 40D <sub>x</sub> | 10    | 0.05  | <b>0.055 <math>\pm</math> 0.001</b> | <b>0.257 <math>\pm</math> 0.005</b> = 0.212 $\pm$ 0.007 + 0.045 $\pm$ 0.010 | 7.85 $\pm$ 0.61                   | 0.059 $\pm$ 0.001                   | 0.320 $\pm$ 0.006 = 0.210 $\pm$ 0.010 + 0.110 $\pm$ 0.013                   | <b>1.44 <math>\pm</math> 0.05</b> |
| Buckling    | 10D <sub>x</sub> | 4     | 0.005 | <b>0.174 <math>\pm</math> 0.094</b> | 1.567 $\pm$ 1.348 = 0.044 $\pm$ 0.021 + 1.523 $\pm$ 1.346                   | 7.97 $\pm$ 1.70                   | 0.327 $\pm$ 0.534                   | <b>0.659 <math>\pm</math> 0.357</b> = 0.656 $\pm$ 0.357 + 0.002 $\pm$ 0.005 | <b>1.03 <math>\pm</math> 0.05</b> |
|             | 10D <sub>x</sub> | 4     | 0.05  | <b>0.304 <math>\pm</math> 0.121</b> | 3.384 $\pm$ 1.378 = 0.203 $\pm$ 0.073 + 3.181 $\pm$ 1.394                   | 4.81 $\pm$ 1.92                   | 0.423 $\pm$ 0.385                   | <b>1.097 <math>\pm</math> 0.369</b> = 1.032 $\pm$ 0.372 + 0.065 $\pm$ 0.028 | <b>1.07 <math>\pm</math> 0.04</b> |
|             | 40D <sub>x</sub> | 4     | 0.005 | <b>0.016 <math>\pm</math> 0.013</b> | <b>0.056 <math>\pm</math> 0.040</b> = 0.025 $\pm$ 0.002 + 0.031 $\pm$ 0.040 | 11.56 $\pm$ 2.43                  | 0.100 $\pm$ 0.138                   | 0.160 $\pm$ 0.046 = 0.153 $\pm$ 0.046 + 0.006 $\pm$ 0.001                   | <b>1.09 <math>\pm</math> 0.02</b> |
|             | 40D <sub>x</sub> | 4     | 0.05  | <b>0.078 <math>\pm</math> 0.024</b> | <b>0.360 <math>\pm</math> 0.109</b> = 0.215 $\pm$ 0.015 + 0.145 $\pm$ 0.108 | 6.24 $\pm$ 2.76                   | 0.188 $\pm$ 0.257                   | 0.485 $\pm$ 0.077 = 0.385 $\pm$ 0.084 + 0.100 $\pm$ 0.014                   | <b>1.09 <math>\pm</math> 0.27</b> |
| Ackley      | 10D <sub>x</sub> | 10    | 0.005 | <b>0.301 <math>\pm</math> 0.023</b> | <b>1.569 <math>\pm</math> 0.173</b> = 1.297 $\pm$ 0.185 + 0.271 $\pm$ 0.271 | <b>1.16 <math>\pm</math> 0.19</b> | 0.379 $\pm$ 0.039                   | 1.879 $\pm$ 0.174 = 1.796 $\pm$ 0.148 + 0.084 $\pm$ 0.096                   | 1.36 $\pm$ 0.04                   |
|             | 10D <sub>x</sub> | 10    | 0.05  | <b>0.310 <math>\pm</math> 0.022</b> | <b>1.616 <math>\pm</math> 0.172</b> = 1.296 $\pm$ 0.178 + 0.321 $\pm$ 0.264 | <b>1.15 <math>\pm</math> 0.13</b> | 0.386 $\pm$ 0.039                   | 1.932 $\pm$ 0.176 = 1.835 $\pm$ 0.149 + 0.097 $\pm$ 0.089                   | 1.38 $\pm$ 0.27                   |
|             | 40D <sub>x</sub> | 10    | 0.005 | 0.213 $\pm$ 0.004                   | 1.073 $\pm$ 0.027 = 0.881 $\pm$ 0.045 + 0.192 $\pm$ 0.047                   | 7.48 $\pm$ 0.75                   | <b>0.209 <math>\pm</math> 0.007</b> | <b>0.990 <math>\pm</math> 0.029</b> = 0.782 $\pm$ 0.035 + 0.208 $\pm$ 0.048 | <b>1.52 <math>\pm</math> 0.05</b> |
|             | 40D <sub>x</sub> | 10    | 0.05  | 0.220 $\pm$ 0.004                   | 1.112 $\pm$ 0.027 = 0.920 $\pm$ 0.046 + 0.192 $\pm$ 0.052                   | 7.85 $\pm$ 0.72                   | <b>0.216 <math>\pm</math> 0.007</b> | <b>1.032 <math>\pm</math> 0.032</b> = 0.810 $\pm$ 0.042 + 0.223 $\pm$ 0.057 | <b>1.52 <math>\pm</math> 0.05</b> |
|             | 10D <sub>x</sub> | 20    | 0.005 | <b>0.316 <math>\pm</math> 0.013</b> | <b>1.614 <math>\pm</math> 0.079</b> = 1.231 $\pm$ 0.086 + 0.383 $\pm$ 0.140 | 3.39 $\pm$ 0.34                   | 0.392 $\pm$ 0.026                   | 1.888 $\pm$ 0.106 = 1.748 $\pm$ 0.115 + 0.140 $\pm$ 0.069                   | <b>1.98 <math>\pm</math> 0.04</b> |
|             | 10D <sub>x</sub> | 20    | 0.05  | <b>0.321 <math>\pm</math> 0.014</b> | <b>1.621 <math>\pm</math> 0.091</b> = 1.234 $\pm$ 0.090 + 0.387 $\pm$ 0.155 | 3.45 $\pm$ 0.34                   | 0.398 $\pm$ 0.026                   | 1.885 $\pm$ 0.102 = 1.751 $\pm$ 0.110 + 0.134 $\pm$ 0.071                   | <b>1.96 <math>\pm</math> 0.04</b> |
|             | 40D <sub>x</sub> | 20    | 0.005 | 0.217 $\pm$ 0.003                   | 1.069 $\pm$ 0.014 = 0.901 $\pm$ 0.026 + 0.168 $\pm$ 0.033                   | 59.99 $\pm$ 8.36                  | <b>0.213 <math>\pm</math> 0.003</b> | <b>1.000 <math>\pm</math> 0.017</b> = 0.837 $\pm$ 0.019 + 0.163 $\pm$ 0.024 | <b>2.36 <math>\pm</math> 0.03</b> |
|             | 40D <sub>x</sub> | 20    | 0.05  | 0.225 $\pm$ 0.003                   | 1.096 $\pm$ 0.014 = 0.921 $\pm$ 0.025 + 0.175 $\pm$ 0.032                   | 60.64 $\pm$ 9.48                  | <b>0.220 <math>\pm</math> 0.003</b> | <b>1.032 <math>\pm</math> 0.017</b> = 0.862 $\pm$ 0.018 + 0.170 $\pm$ 0.023 | <b>2.37 <math>\pm</math> 0.02</b> |
|             | 10D <sub>x</sub> | 40    | 0.005 | <b>0.298 <math>\pm</math> 0.011</b> | <b>1.495 <math>\pm</math> 0.047</b> = 1.265 $\pm$ 0.039 + 0.230 $\pm$ 0.062 | 18.04 $\pm$ 3.29                  | 0.402 $\pm$ 0.018                   | 1.945 $\pm$ 0.071 = 1.793 $\pm$ 0.061 + 0.152 $\pm$ 0.053                   | <b>3.20 <math>\pm</math> 0.06</b> |
|             | 10D <sub>x</sub> | 40    | 0.05  | <b>0.307 <math>\pm</math> 0.017</b> | <b>1.531 <math>\pm</math> 0.080</b> = 1.281 $\pm$ 0.053 + 0.250 $\pm$ 0.068 | 17.76 $\pm$ 3.08                  | 0.409 $\pm$ 0.018                   | 1.977 $\pm$ 0.074 = 1.813 $\pm$ 0.061 + 0.164 $\pm$ 0.056                   | <b>3.26 <math>\pm</math> 0.06</b> |
|             | 40D <sub>x</sub> | 40    | 0.005 | 0.243 $\pm$ 0.007                   | 1.182 $\pm$ 0.042 = 0.922 $\pm$ 0.018 + 0.260 $\pm$ 0.051                   | 468.06 $\pm$ 68.10                | <b>0.231 <math>\pm</math> 0.005</b> | <b>1.112 <math>\pm</math> 0.024</b> = 1.006 $\pm$ 0.018 + 0.106 $\pm$ 0.018 | <b>4.85 <math>\pm</math> 0.03</b> |
|             | 40D <sub>x</sub> | 40    | 0.05  | 0.251 $\pm$ 0.006                   | 1.202 $\pm$ 0.035 = 0.944 $\pm$ 0.017 + 0.259 $\pm$ 0.046                   | 476.18 $\pm$ 90.01                | <b>0.236 <math>\pm</math> 0.005</b> | <b>1.135 <math>\pm</math> 0.023</b> = 1.028 $\pm$ 0.019 + 0.107 $\pm$ 0.016 | <b>4.85 <math>\pm</math> 0.04</b> |
| Dixon Price | 10D <sub>x</sub> | 10    | 0.005 | 0.442 $\pm$ 0.028                   | 2.037 $\pm$ 0.263 = 1.539 $\pm$ 0.155 + 0.498 $\pm$ 0.342                   | <b>1.29 <math>\pm</math> 0.19</b> | <b>0.427 <math>\pm</math> 0.072</b> | <b>2.007 <math>\pm</math> 0.318</b> = 1.855 $\pm$ 0.142 + 0.153 $\pm$ 0.275 | 1.37 $\pm$ 0.04                   |
|             | 10D <sub>x</sub> | 10    | 0.05  | 0.453 $\pm$ 0.051                   | 2.085 $\pm$ 0.497 = 1.521 $\pm$ 0.169 + 0.564 $\pm$ 0.560                   | <b>1.32 <math>\pm</math> 0.13</b> | <b>0.428 <math>\pm</math> 0.074</b> | <b>2.019 <math>\pm</math> 0.322</b> = 1.879 $\pm$ 0.145 + 0.140 $\pm$ 0.285 | 1.40 $\pm$ 0.06                   |
|             | 40D <sub>x</sub> | 10    | 0.005 | 0.279 $\pm$ 0.019                   | 1.287 $\pm$ 0.099 = 1.049 $\pm$ 0.132 + 0.237 $\pm$ 0.074                   | 11.43 $\pm$ 1.09                  | <b>0.162 <math>\pm</math> 0.018</b> | <b>0.839 <math>\pm</math> 0.076</b> = 0.790 $\pm$ 0.059 + 0.049 $\pm$ 0.031 | <b>1.53 <math>\pm</math> 0.26</b> |
|             | 40D <sub>x</sub> | 10    | 0.05  | 0.288 $\pm$ 0.018                   | 1.325 $\pm$ 0.102 = 1.080 $\pm$ 0.148 + 0.245 $\pm$ 0.061                   | 11.27 $\pm$ 1.04                  | <b>0.178 <math>\pm</math> 0.016</b> | <b>0.902 <math>\pm</math> 0.068</b> = 0.826 $\pm$ 0.057 + 0.076 $\pm$ 0.030 | <b>1.52 <math>\pm</math> 0.07</b> |
|             | 10D <sub>x</sub> | 20    | 0.005 | 0.474 $\pm$ 0.027                   | 2.286 $\pm$ 0.274 = 1.553 $\pm$ 0.126 + 0.733 $\pm$ 0.367                   | 4.83 $\pm$ 0.70                   | <b>0.470 <math>\pm</math> 0.028</b> | <b>2.161 <math>\pm</math> 0.127</b> = 1.913 $\pm$ 0.121 + 0.248 $\pm$ 0.088 | <b>1.94 <math>\pm</math> 0.03</b> |
|             | 10D <sub>x</sub> | 20    | 0.05  | <b>0.466 <math>\pm</math> 0.025</b> | 2.227 $\pm$ 0.272 = 1.569 $\pm$ 0.124 + 0.658 $\pm$ 0.359                   | 4.47 $\pm$ 0.51                   | 0.473 $\pm$ 0.028                   | <b>2.183 <math>\pm</math> 0.126</b> = 1.916 $\pm$ 0.129 + 0.266 $\pm$ 0.085 | <b>1.94 <math>\pm</math> 0.03</b> |
|             | 40D <sub>x</sub> | 20    | 0.005 | 0.334 $\pm$ 0.006                   | 1.558 $\pm$ 0.031 = 1.300 $\pm$ 0.044 + 0.258 $\pm$ 0.052                   | 100.19 $\pm$ 8.90                 | <b>0.195 <math>\pm</math> 0.009</b> | <b>0.977 <math>\pm</math> 0.049</b> = 0.911 $\pm$ 0.051 + 0.066 $\pm$ 0.018 | <b>2.36 <math>\pm</math> 0.27</b> |
|             | 40D <sub>x</sub> | 20    | 0.05  | 0.339 $\pm$ 0.005                   | 1.561 $\pm$ 0.031 = 1.329 $\pm$ 0.076 + 0.233 $\pm$ 0.074                   | 96.79 $\pm$ 9.05                  | <b>0.201 <math>\pm</math> 0.008</b> | <b>1.014 <math>\pm</math> 0.045</b> = 0.935 $\pm$ 0.051 + 0.079 $\pm$ 0.021 | <b>2.36 <math>\pm</math> 0.02</b> |
|             | 10D <sub>x</sub> | 40    | 0.005 | <b>0.455 <math>\pm</math> 0.025</b> | <b>2.183 <math>\pm</math> 0.153</b> = 1.618 $\pm$ 0.076 + 0.565 $\pm$ 0.155 | 31.03 $\pm$ 5.24                  | 0.480 $\pm$ 0.033                   | 2.324 $\pm$ 0.135 = 2.007 $\pm$ 0.105 + 0.317 $\pm$ 0.107                   | <b>3.30 <math>\pm</math> 0.05</b> |
|             | 10D <sub>x</sub> | 40    | 0.05  | <b>0.470 <math>\pm</math> 0.022</b> | <b>2.240 <math>\pm</math> 0.135</b> = 1.635 $\pm$ 0.076 + 0.605 $\pm$ 0.146 | 32.26 $\pm$ 5.40                  | 0.484 $\pm$ 0.032                   | 2.337 $\pm$ 0.138 = 2.010 $\pm$ 0.104 + 0.327 $\pm$ 0.109                   | <b>3.30 <math>\pm</math> 0.06</b> |
|             | 40D <sub>x</sub> | 40    | 0.005 | 0.431 $\pm$ 0.047                   | 2.086 $\pm$ 0.336 = 1.481 $\pm$ 0.082 + 0.605 $\pm$ 0.284                   | 878.56 $\pm$ 130.06               | <b>0.231 <math>\pm</math> 0.009</b> | <b>1.140 <math>\pm</math> 0.034</b> = 1.066 $\pm$ 0.029 + 0.075 $\pm$ 0.018 | <b>4.85 <math>\pm</math> 0.27</b> |
|             | 40D <sub>x</sub> | 40    | 0.05  | 0.433 $\pm$ 0.044                   | 2.040 $\pm$ 0.300 = 1.491 $\pm$ 0.072 + 0.549 $\pm$ 0.260                   | 894.63 $\pm$ 106.36               | <b>0.237 <math>\pm</math> 0.009</b> | <b>1.166 <math>\pm</math> 0.034</b> = 1.087 $\pm$ 0.029 + 0.079 $\pm$ 0.019 | <b>4.86 <math>\pm</math> 0.03</b> |
| Griewank    | 10D <sub>x</sub> | 10    | 0.005 | <b>0.011 <math>\pm</math> 0.051</b> | <b>0.111 <math>\pm</math> 0.388</b> = 0.111 $\pm$ 0.388 + 0.000 $\pm$ 0.000 | <b>1.25 <math>\pm</math> 0.13</b> | 0.511 $\pm$ 0.072                   | 2.458 $\pm$ 0.314 = 2.267 $\pm$ 0.193 + 0.190 $\pm$ 0.266                   | 1.38 $\pm$ 0.03                   |
|             | 10D <sub>x</sub> | 10    | 0.05  | <b>0.109 <math>\pm</math> 0.097</b> | <b>0.529 <math>\pm</math> 0.590</b> = 0.482 $\pm$ 0.531 + 0.047 $\pm$ 0.162 | <b>1.21 <math>\pm</math> 0.16</b> | 0.520 $\pm$ 0.074                   | 2.483 $\pm$ 0.321 = 2.277 $\pm$ 0.199 + 0.207 $\pm$ 0.275                   | 1.35 $\pm$ 0.12                   |
|             | 40D <sub>x</sub> | 10    | 0.005 | <b>0.007 <math>\pm</math> 0.000</b> | <b>0.033 <math>\pm</math> 0.001</b> = 0.028 $\pm$ 0.001 + 0.005 $\pm$ 0.001 | 11.93 $\pm$ 0.99                  | 0.099 $\pm$ 0.006                   | 0.470 $\pm$ 0.028 = 0.445 $\pm$ 0.023 + 0.025 $\pm$ 0.011                   | <b>1.42 <math>\pm</math> 0.04</b> |
|             | 40D <sub>x</sub> | 10    | 0.05  | <b>0.056 <math>\pm</math> 0.001</b> | <b>0.262 <math>\pm</math> 0.003</b> = 0.219 $\pm$ 0.007 + 0.042 $\pm$ 0.008 | 11.61 $\pm$ 0.88                  | 0.117 $\pm$ 0.005                   | 0.558 $\pm$ 0.024 = 0.483 $\pm$ 0.022 + 0.075 $\pm$ 0.013                   | <b>1.43 <math>\pm</math> 0.04</b> |
|             | 10D <sub>x</sub> | 20    | 0.005 | <b>0.139 <math>\pm</math> 0.020</b> | <b>1.009 <math>\pm</math> 0.008</b> = 1.009 $\pm$ 0.009 + 0.000 $\pm$ 0.005 | 3.83 $\pm$ 0.28                   | 0.516 $\pm$ 0.073                   | 2.425 $\pm$ 0.331 = 2.175 $\pm$ 0.174 + 0.250 $\pm$ 0.202                   | <b>1.96 <math>\pm</math> 0.03</b> |
|             | 10D <sub>x</sub> | 20    | 0.05  | <b>0.174 <math>\pm</math> 0.020</b> | <b>1.065 <math>\pm</math> 0.020</b> = 1.060 $\pm$ 0.023 + 0.005 $\pm$ 0.020 | 3.85 $\pm$ 0.41                   | 0.522 $\pm$ 0.075                   | 2.434 $\pm$ 0.340 = 2.185 $\pm$ 0.179 + 0.249 $\pm$ 0.203                   | <b>1.98 <math>\pm</math> 0.27</b> |
|             | 40D <sub>x</sub> | 20    | 0.005 | <b>0.008 <math>\pm</math> 0.000</b> | <b>0.040 <math>\pm</math> 0.001</b> = 0.035 $\pm$ 0.001 + 0.005 $\pm$ 0.001 | 88.39 $\pm$ 8.93                  | 0.140 $\pm$ 0.006                   | 0.644 $\pm$ 0.022 = 0.600 $\pm$ 0.016 + 0.044 $\pm$ 0.013                   | <b>2.37 <math>\pm</math> 0.03</b> |
|             | 40D <sub>x</sub> | 20    | 0.05  | <b>0.061 <math>\pm</math> 0.001</b> | <b>0.282 <math>\pm</math> 0.006</b> = 0.242 $\pm$ 0.006 + 0.040 $\pm$ 0.007 | 92.60 $\pm$ 10.48                 | 0.148 $\pm$ 0.006                   | 0.698 $\pm$ 0.020 = 0.632 $\pm$ 0.017 + 0.066 $\pm$ 0.015                   | <b>2.35 <math>\pm</math> 0.03</b> |
|             | 10D <sub>x</sub> | 40    | 0.005 | <b>0.255 <math>\pm</math> 0.045</b> | <b>1.552 <math>\pm</math> 0.068</b> = 1.543 $\pm$ 0.032 + 0.009 $\pm$ 0.040 | 23.15 $\pm$ 3.12                  | 0.548 $\pm$ 0.043                   | 2.544 $\pm$ 0.171 = 2.206 $\pm$ 0.120 + 0.338 $\pm$ 0.094                   | <b>3.25 <math>\pm</math> 0.06</b> |
|             | 10D <sub>x</sub> | 40    | 0.05  | <b>0.257 <math>\pm</math> 0.038</b> | <b>1.562 <math>\pm</math> 0.064</b> = 1.553 $\pm$ 0.033 + 0.009 $\pm$ 0.033 | 23.26 $\pm$ 3.52                  | 0.552 $\pm$ 0.043                   | 2.562 $\pm$ 0.171 = 2.231 $\pm$ 0.120 + 0.331 $\pm$ 0.088                   | <b>3.26 <math>\pm</math> 0.06</b> |
|             | 40D <sub>x</sub> | 40    | 0.005 | <b>0.008 <math>\pm</math> 0.000</b> | <b>0.049 <math>\pm</math> 0.001</b> = 0.047 $\pm$ 0.001 + 0.002 $\pm$ 0.000 | 894.56 $\pm$ 203.48               | 0.173 $\pm$ 0.005                   | 0.968 $\pm$ 0.028 = 0.900 $\pm$ 0.023 + 0.067 $\pm$ 0.020                   | <b>4.85 <math>\pm</math> 0.26</b> |
|             | 40D <sub>x</sub> | 40    | 0.05  | <b>0.075 <math>\pm</math> 0.003</b> | <b>0.374 <math>\pm</math> 0.012</b> = 0.295 $\pm$ 0.015 + 0.079 $\pm$ 0.013 | 922.65 $\pm$ 180.50               | 0.181 $\pm$ 0.004                   | 0.996 $\pm$ 0.028 = 0.927 $\pm$ 0.022 + 0.069 $\pm$ 0.021                   | <b>4.84 <math>\</math></b>        |